

AP EAMCET AGRICULTURE AND MEDICINE

Date : 23rd April 2019 Number of Questions :160

Shift – II

1. Match the following

List - I	List -II	List-III
A) Clara	i) Selaginella	I) Having long and dwarf branches
B) Algin	ii) Pyrenoid	II) Salgassum
C) Microphylls	iii) Living fossil	III) Heterosporous
D) Ginkgo	iv) Water holding substances	IV) Cellulose and pectins

- A. A B C D
 ii, IV i, III iv, II iii, I
 B. A B C D

	ii, II	iv, I	iii, IV	i, III
C.	A	B	C	D
	iii, II	i, I	iv, III	ii, IV
D.	A	B	C	D
	ii, IV	iv, II	i, III	iii, I

2. Match the following

List – I

A) Protostele

B) Siphonostele

C) Dictyostele

D) Solonostele

A B C D

A. I II III IV

A B C D

B. II I IV III

A B C D

C. III IV II I

A B C D

D. III II I IV

List – II

I) dissected siphonostele with leaf gap

II) Bisected siphonostele with over lapping leaf gaps

III) Xylem surrounded by phloem

IV) Medullated protostele

3. Study the following lists and choose the correct match

List - I

A) Tracheophytic.
Archegoniate.

B) Tracheophytic.

C) Atracheophytic.
Archegoniate

D) Tracheophytic.
archegoniate

List - II

I) Angiosperms
Embryophytic.
Phanerogams

II) Gymnosperms
Embryophytic.
Cryptogams

III) Algae
Embryophytic.
Cryptogams

IV) Bryophytes Non-
Embryophytic.
Phanerogams

V) Pteridophytes

A B C D

A. II V IV III

A B C D

B. I III V II

A B C D

C. II V IV I

A B C D

D. II IV III I

4. Fill up the Blanks with suitable words

The gametophytes in gymnosperms are _____
but Cycas shows _____ because of
presence of multiciliate male gamete. The retained
multicellular female gametophyte is also called
_____. In this thick cuticle and
_____ will be useful to reduce water loss.

A Sunken stomata

B. Zooidogamy

C. Endosperm

D. Dependent

A. D,B,C,A

B. B,C,D,A

C. B,D,A,C

D. D,A,C,B

5. Match the following

List - I

A) False septum
in Ovary

B) Carrot root

List - II

i) Fig

ii) Banana

List -III

I) Storage of
food and
respiration

II) Length of

C) Sucker

iii) Mustard

Stamen
filament
varies
III) Sub
aerial
stem
modificat
ion

D) Hypanthoditun

iv) Absorption

IV) Blastophaga

The correct answer is

- | | A | B | C | D |
|----|-------|---------|---------|------|
| A. | ii,II | iii,III | i,IV | iv,I |
| | A | B | C | D |
| B. | iii,I | ii,III | i,IV | iv,I |
| | A | B | C | D |
| C. | i,II | iv,IV | iii,III | ii,I |
| | A | B | C | D |
| D. | iii,I | iv,I | ii,III | i,IV |

6. Match the following

- | List - I | List - II | List -III |
|----------------------|------------------|--------------|
| A) Acacia | i) Node of stein | I) Mericarp |
| B) Dichasial
cyme | ii) False septum | II) amiaceae |

- C) Lateral out growth iii) Monochasial cyme III) Photosynthesis
D) A, gemone iv) Phyllode IV) Parietal placentation

The correct answer is

- | | A | B | C | D |
|----|--------|---------|----------|----------|
| A. | iv, I | iii, II | i, III | ii, IV |
| B. | iii, I | ii, II | i, IV | iv, III |
| C. | iv, I | ii, II | iii, III | i, IV |
| D. | iv, II | ii, I | i, IV | iii, III |

7. Choose the correct statement/s
- A In majority of algae formation of zygote occurs in female plant
- B. Cladophora produces identical gametes
- C. In Bryophytes and Pteridophytes the ratio of male gametes to female gametes will be several thousands
- D. In Mangroves seeds germinate while in mother plant to increase environmental stress

- A. A, B
B. A, B, C
C. B, C
D. A, B, C, D

8. Fill up the blanks with suitable words

The ability of the pistil to recognise pollen is dependent on _____ components and _____ guide the entry of pollentube. This study leads to help _____ in getting _____ even in _____

- A. Chemicals
- B. Plant breeders
- C. hybrids
- D. Incompatible pollination
- E. Synergids

- A. A, B, C, D, E
- B. A, E, B, C, D
- C. E, A, D, C, B
- D. A, C, B, D, E

9. Assertion (A) : Unlike coconut. groundnut consume endosperm completely during embryo development

Reason (R) : Though the embryogeny is similar in groundnut and coconut. perisperm is seen in coconut

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)

- B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

10. Choose the wrong statement :

- A. In hydrophilly the pollengrains are covered by mucilagenous layer and in water hyacinth pollination takes place by insects
 - B. Nector and pollen are the rewards for animals in pollination but not in Yucca and Ammophallus
 - C. Autogamy can be prevented by herkogamy and cliestogamous flowers
 - D. Wind pollinated flowers will have multi ovule condition for more seed production
- A. A, B
 - B. B, C
 - C. B, C
 - D. A, D

11. Choose the correct combination

List - I

List -II

List -III

A) Raceme

i) Cup strapped
thallamus

I) &Irk Wk.!:

B) Mathematical ii) Codes

II) Use of

model		computers
C) Trimerous	iii) Species plantam	III) Artificial System
D) Historia plantarum	iv) Capsule	IV) Alliwn

The correct answer is

- A. A, D
- B. B,D
- C. A,C
- D. A,B

12. Assertion (A) : Cis and Trans faces of the golgi associated with glyco proteins and glyco lipids
Reason (R) : Golgi apparaus is in close association with rough Endoplasmic reticulum

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

13. Choose the correct combination

List - I	List - II	List -III
A) Radial spikes	i) Sealing of DNA	I) Axoneme

- | | | |
|---------------------------|-----------------------------|--------------------------|
| B) H1 Histone | ii) Positive charge histone | II) Lies outside of core |
| C) Negatively charged DNA | iii) Linkers | III) DNA package |
| D) Nucleolus | iv) No membrane | IV) r-RNA synthesis |

The correct answer is

- A. A, C
- B. B, D
- C. B, C
- D. A, D

14. Choose the correct statements

- A. Acid soluble fractions have more than one thousand daltons
- B. Common acid used in the living cell compound analysis is Hydrochloric acid
- C. Number of carbons in Arachidonic acid is 20
- D. Serine is the hydroxy methyl group amino acid

The correct answer is

- A. A, B
- B. B, C
- C. C, D

D. A, D

15. Find the wrong statement

- A. Polymers of polysaccharides, fats and proteins are synthesised from monomers by ionic bonding
- B. Molecules with polarheads and hydrophobic tails are present in cell membrane
- C. Left end of protein chain is N-terminal end
- D. Right and Left handed helices are observed in protein

The correct answer is

- A. A, B
- B. B, C
- C. C, D
- D. A, D

16. Fill up the blanks with suitable words

In Meiosis, chromosomes start _____ which is called _____ thus a _____ of chromatids. Later the nonsister chromatids will _____ undergo leading to genetic material

A B C D

Pairing Synapsis Exchange Tetrad Crossing over

- A. $A \rightarrow C \rightarrow D \rightarrow B \rightarrow E$
- B. $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C$
- C. $B \rightarrow A \rightarrow C \rightarrow D \rightarrow E$
- D. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

17. Assertion (A) : Chromosomes are inactive and not transcribed to messenger RNA in phase of apparent division

Reason (R) Cells remains metabolically active but no longer divide in quiescent stage

- A. Both (A) and (R) are true. and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

18. Sclerenchyma gives mechanical support in

- A. Seed coat of Legume. Fruit pulp of Guava. Leaf of Tea
- B. Seed coat of Sapota. Leaf of Guava. Pulp of Legume
- C. Seed coat of Guava. Leaf of Legumes Fruit pulp of Tea
- D. Leaf of Pear. Seed coat of guava. Pulp of Legumes

19. Epidermal hairs of shoot system I. Unicellular or Multicellular II. Multicellular. branched or unbranched III. Secretary in function IV. Absorbs water V Prevent transpiratory water loss IL ItL V
- II, III, V
 - I, II, III, V
 - II, III, IV
 - I. IV V
20. Plants which show given characters respectively
- Long petiolated leaves. Rooted hydrophytes with floating leaves
 - Succulent roots. water in the form of mucilage
 - Submerged and not rooted hydrophyte
 - Grows in saline soils, respiratory roots
- Nt.mphaea, Salvinia, Tipha, Vanda
 - Victoria regia, Asparagus, Utricularia, Rhizophora
 - Sagittaria, Pistia, Hydrila, Vallisneria
 - Victoria regia, Asparagus, Salvinia, Rhizophora
21. Choose the correct statements

A. Porins allow molecules of small proteins present in membrane of Mitochondria, Chloroplast and bacteria

B. When a living cell kept in hypertonic solution, the water potential is equal to solute potential C. Carrier proteins will allow all substances across the membrane where as pumps transport large molecules in a passive method

D. Bulk flow can be achieved through diffusion

The correct answer is

A. A, B

B. C, D

C. A, D

D. B, C

22. Choose the correct combination

List - I

List -II

List-III

A) Mo

i) Delay in flowering

I) And transported through Xylem

B) Amides

ii) Contain more Nitrogen

II) Whiptail in Cauliflower

C) Hydroponics

iii) Commercial Production of vegetables

III) Seedless Cucumber

D) Pseudomonas iv) Denitrification IV) Reduced to N_2

- A. A, B
- B. A, C
- C. C, D
- D. B, D

23. Arrange the sequences in enzyme action

- A) Product releases and free enzyme again binds to another substrate molecule
- B) Enzyme to alter its shape
- C) Substrate binds to active site of the enzyme
- D) Fits in to the active site E. Formation of enzyme product complex

The correct answer is

- A. A, C, D, B, E
- B. A, B, D, C, E
- C. C, B, D, E, A
- D. C, D, B, E, A

24. First reaction in electron transport of Photosynthesis

- A. Photosystem II Chl a absorb light of 680 nm and transport e^- to cytochrome
- B. Photosystem I Chl a absorb light of 700 nm and transport e^- to pheophytin

- C. Photosystem II Chl a absorb light of 680 nm and excite e from atomic nucleus to farther orbit
- D. Photo system I Chl a absorb light of 700 tun and transport e to cytochronie

25. Choose the correct statement

- A) Difference between C₄ and CAM pathway is the CO₂ fixation and Calvin cycle are separated in space and time respectively
- B) Aldolase combines two 3-carbon compounds as well as one 4-carbon and one 7-carbon compound
- C) Based on the labelled carbon studies CO₂ fixation products were identified
- D) Stoma lamella and Grana lamella are identical in the distribution of the photosystems

The correct answer is

- A. A, C
- B. B, C
- C. A, D
- D. B, D

26. Choose the correct statement

- A) Reduced ubiquinone is oxidized with transfer of electrons to cytochrome C via complex III

- B) Intermediates in the respiratory pathway are not utilized for synthesis of other compounds
C) Fatty acid will be broken down to acetyl CoA before entering the respiratory pathway
D) The energy released by oxidation in respiration is not used directly
- A. A, B, C
B. B, C, D
C. A, C, D
D. A, B, C, D

27. Identify the wrong statements in the following

- A) Winter varieties when planted in spring normally come to flower before growing season
B) In short day plants above critical photoperiod flowering will be initiated
C) Kinetin has specific effect on cytokinesis which is a modified purine
D) Auxins are used as herbicides

The correct answer is

- A. A, B
B. B, C
C. A, C
D. A, D

28. Fill up blanks with suitable words

In a graph of length of organ against time, which is a typical _____ and expressed as _____

However, in a graph of growth against time

_____ will appear with expression of

_____. Where _____ is base of natural logarithm and also 'r' as _____.

^A Sigmoid curve ^B $W_1 = W_0 e^{rt}$ ^C $L_t = L_0 + rt$ ^D Linear ^E e ^F Efficiency index

A. D, C, A, B, E, F

B. A, B, C, D, E, F

C. C, B, A, D, F, E

D. A, B, E, F, C, D

29. Match the following

List - I	List - II	List-III
A) Discovery of DNA as genetic material	i) Transforming principle	I) Conjugation
B) Chromium	ii) Derive energy from oxidation of inorganic substance	II) Obtain carbon from atmosphere CO ₂

C) Conservative process	iii) Capture light energy	III) S. pneumoniae
D) Beggiotoa	iv) Donor retain a copy of genetic material	IV) Obtain carbon from CO ₂

The correct answer is

- | | | | | |
|----|---------|---------|---------|---------|
| | A | B | C | D |
| A. | ii, III | i, I | iii, II | iv, IV |
| | A | B | C | D |
| B. | i, III | iii, II | iv, I | ii, IV |
| | A | B | C | D |
| C. | i, II | ii, I | iv, III | iii, IV |
| | A | B | C | D |
| D. | iv, III | i, II | iii, I | ii, IV |

30. Assertion (A) : DNA from Bacteriophage head passes through tail core through plasma membrane and enters bacterial cell

Reason (R) : Plasma membrane of the host cell gets dissolved due to lysozyme and releases the new virions

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

31. Choose the wrong statement.

- A) A single gene product may produce more than one effect this it relates to pleiotropy
- B) Dominance is an autonomous feature of a gene or the product
- C) The frequency of recombination between gene pairs on the same chromosome as a measure of distance between those genes
- D) Breeder cannot select the mutations of the desirable types as it has less variability

The correct answer is

- A. A, B
- B. B, C
- C. A, D
- D. B, D

32. Choose the correct statement

A) DNA is chemically more reactive and structurally stable compared to RNA

B) Catalytic RNA is known as Ribozymes

C) DNA can directly code for the synthesis of protein. where as RNA dependent on DNA for protein synthesis

D) Presence of 5-methyl uracil in the DNA also confers stability

The correct answer is

A. B, D

B. B, C

C. A, B

D. C, D

33. Assertion (A) : During transcription both the strands of DNA are copied

Reason (R) : If both strands act as template DNA would code for RNA molecule with different sequences

A. Both (A) and (R) are correct and (R) is the correct explanation of (A)

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C. (A) is correct but (R) is not correct

D. (A) is not correct but (R) is correct

34. Match the following

List -I	List - II	List -III
A) Tailing	i) 28s. 18s. 5.8s	I) Transported out of nucleus
B) r-RNA	ii) Addition of methyl guanosine triphosphate	II) 3' end
C) Capping	iii) m-RNA	III) Transcribes by RNA polymerase
D) Processed hn RNA	iv) Addition of Adenylate Residue	IV) 5' end of hn RNA

The correct answer is

- | | | | | |
|----|---------|--------|----------|---------|
| | A | B | C | D |
| A. | iv, II | i, III | ii, IV | iii, I |
| | A | B | C | D |
| B. | iv, II | ii, IV | iii, III | i, I |
| | A | B | C | D |
| C. | iv, III | ii, IV | i, I | iii, II |
| | A | B | C | D |

D. ii, I i, III iii, IV iv, II

35. Find out the series of genes in the transcription unit of the following statements

I. Located towards 3' end of the structural gene

II. Flanked by promoter and terminator

III. Located towards 3' end of the coding strand

A. Coding strand, Template strand, Structural gene

B. Promoter, Structural gene, Terminator

C. Terminator, Coding strand. Structural gene

D. Terminator. Structural gene, Promoter

36. Find the wrong statements

A) To cut DNA with restriction enzyme it needs to be pure form. free from RNA. Protein. Polysaccharides and lipids

B) RNA can be removed by protease enzyme

C) DNA separated as suspension can be removed by spooling

D) DNA can be precipitated by adding warm ethanol

The correct answer is

A. A, B

B. C, D

C. A, C

D. B, D

37. Choose the correct statements

- A) Early detection of pathogen presence can be done by PCR
- B) Biochemical products produced by bacteria are inferior to products produced in transgenic plants
- C) Validity of the GM Research will be controlled by Genetic Engineering Approval Committee
- D) Use of bio resources by multinational companies with proper authorization is not possible

The correct answer is

- A. B, C
- B. A, B
- C. B, D
- D. A, C

38. Arrange the steps to be followed in breeding techniques

- A) Creating the plants as pure lines
- B. Progeny of the hybrids to be tested for the homozygosity for generations
- C) Evaluation by growing in research field. fanner field and then for three growing seasons in several locations
- D) Evaluation with best available local crop cultivars

E) Pollen grain of desirable male plant to be collected and placed on the stigma of the selected female plant

F) Effective exploitation of natural genes available in the population

A. F B A E D C

B. B A C D E F

C. F A E B C D

D. A F E B C D

39. Match the following

List - I	List-II	List-III
A) Hairy leaves	i) Animal manure and sewage	I. Potato
B) Spirulina	ii) Bacterial Blight	II. Jassids in cotton
C) Apical meristem	iii) Resistance to pests	III. Cow pea
D) Pusa komal	iv) Disease Resistance	IV Rich with protein

The correct answer is

A	B	C	D
A. i. II	ii, III	iii, IV	iii, I
A	B	C	D

- B. iii. II iv. IV i, I ii. III
 A B C D
- C. i. III ii, II iv: IV iii, I
 A B C D
- D. iii. II i. IV iv, I ii. III

40. Match the following

List - I	List - II	List-III
A) Wine	i) Useful aerobic microbes	I) Beverage
B) Flocs	ii) Without distillation	II) Sedimented flocks
C) Ethanol	iii) Brewer's yeast	III) Associated with fungal filament
D) Activated sludge	iv) Reduced BOD	IV) Fermentation of Malted cereals

The correct answer is

- A B C D

- | | | | | |
|----|---------|----------|----------|--------|
| A. | i, I | ii, II | iii, III | iv, IV |
| | A | B | C | D |
| B. | iv, II | iii, III | ii, I | i, IV |
| | A | B | C | D |
| C. | ii, III | iii, IV | i, II | iv, I |
| | A | B | C | D |
| D. | ii, I | i, III | iii, IV | iv, II |

41. Assertion (A) : Species is an evolutionary unit
Reason (R) : Individuals of a species have similar structure and functional characteristics
- Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (A) is correct but (R) is not correct
 - (A) is not correct but (R) is correct
42. Assertion (A) : Mammary glands are apocrine glands
Reason (R) : In apocrine glands, the entire gland cell disintegrates to discharge the contents
- Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)

- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

43. Study the following statements

- I) Stereocilia are long. non motile cilia like processes and are found in epididymis. internal ear etc.
- II) Os cordis is a cartilage bone
- III) Adjacent myocardial cells are joined by gap junctions
- IV) Nissil bodies represent the smooth endoplasmic reticulum

Identify the correct statements

- A. I, IV
- B. I, III
- C. II, IV
- D. III, IV

44. Match the following

List - I

- A) Golgonia
- B) Trichiuris
- C) Astacus
- D) Lepidoplem-us

List - II

- I) Cmstacea
- II) Polyplacophora
- III) Actinozoa
- IV) Scaphopoda
- V) Aphasmidia

The correct answer is

- A B C D
- A. III IV I II
- A B C D
- B. I III II V
- A B C D
- C. III V I II
- A B C D
- D. IV III II I

45. Study the following statements

- I) Digestion in sponges is extra cellular
- II) Primary function of flame cells in flatworms is osmoregulation
- III) Clitellum is absent in unisexual annelids
- IV) Dipleurula larva is considered as the ancestor of echinodermns

Among the above, correct statements are

- A. I, II, III
- B. I, III, IV
- C. I, II, IV
- D. II, III, IV

46. Renal portal system is absent in

- A. Fishes and Amphibians
- B. Cyclostomes and Fishes

C. Reptiles and Birds

D. Cyclostomes and Mammals

47. Study the following table

S. No.	Group	Characters	Example
I	Urochordata	Tunicin. ventral heart with reverse flow of blood. bisexual. tadpole larva	Doliolum
II	Cephalochordate	Solenocytes. ciliary feeding. presence of atrium. closed circulation without heart. gonads with gonoducts	Branchistoma

III	Vertebrata	Paired fins/limbs. ventral heart. kidneys. hepatic portal system	Bufo
-----	------------	---	------

From the above, the correct combinations are

- A. I, II
- B. I, III
- C. II, III
- D. II only

48. Match the following

List – I

List - II

- | | |
|----------------|-----------------|
| A) Isogamy | I) Plasmodium |
| B) Misogamy | II) Vorticella |
| C) Hologamy | III) Astasia |
| D) Conjugation | IV) Monocystis |
| | V) Trichonympha |

The correct answer is

A B C D

- A. IV I III II
- A B C D

- B. IV I V II
A B C D
- C. II III I IV
A B C D
- D. I II IV V

49. Tashkent ulcers are caused by

- A. Leishmania tropica
B. Leishmania donovoni
C. Trypanosoma gambiense
D. blucher eria bancrofti

50. Assertion (A) : Entamoeba histolytica is an obligate anaerobe

Reason (R) : It lacks mitochondria

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
C. (A) is correct but (R) is not correct
D. (A) is not correct but (R) is correct

51. Match the following

List - I

List -II

- | | |
|--------------------------|------------------------------|
| A) Plasmodium malariae | I) Malignant tertian malaria |
| B) Plasmodium falciparum | II) Mild tertian malaria |
| C) Plasmodium vivax | III) Quartan malaria |
| D) Plasmodium ovale | IV) Dun bar fever |
| | V) Benign tertian malaria |

The correct answer is

- A B C D
- A. III IV V II
- A B C D
- B. IV I V II
- A B C D
- C. III I V II
- A B C D
- D. II V I III

52. Read the following statements and pick up the correct one

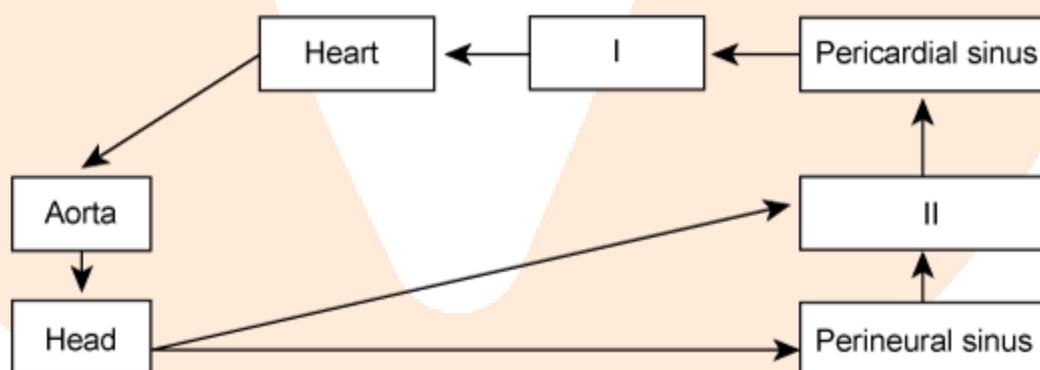
- A. Rhabditiform larva of Ascaris undergoes 4th moulting in alveoli of lungs of man
- B. Inflammation of lymph glands due to blucher eria is called lymphangitis

- C. Cocaine is obtained by the acetylation of morphine
- D. Excessive dosage of cocaine causes hallucinations

53. These cells of corpora adiposa of cockroach store food materials

- A. Mycetocytes
- B. Trophocytes
- C. Oenocytes
- D. Urate cells

54. Study the following regarding blood circulation in cockroach and identify I and II



- A. I - Ostia: II - Perivisceral sinus
- B. I - Haemocoel; II - Ostia
- C. I - Perivisceral sinus: II-Ostia
- D. I -Atria: II - Perivisceral sinus

55. Oriented locomotor movement of an organism towards or away from light source is called

- A. Photokinesis
- B. Phototropism
- C. Phototaxis
- D. Photoperiodism

56. Assertion (A) : Daphnia exhibits cyclomorphosis

Reason (R) : It is an adaptation for Daphnia to lead benthic life

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

57. Match the following

List - I

- A) Metabolic rate
- B) Body size
- C) Size of body parts
- D) Energy flow

List - II

- I) Jordon's rule
- II) Allen's rule
- III) 10 percent law
- IV) Van't Hoff's rule

V) Bergmann's rule

The correct answer is

- A B C D
A. IV I II III
A B C D
B. IV V II III
A B C D
C. III II V IV
A B C D
D. IV V II I

58. Study the following statements

- I) Camels are partial regulators or partial conformers
II) In limnetic zone of a lake. decomposers are almost absent
III) Ten percent law for the transfer of energy from one trophic level to the next was proposed by Elton
IV) The natural reservoir of phosphorus is sea
V) In case of parasitic food chain, the pyramid of number is inverted

Among the above, the incorrect statements are

- A. III V
B. I, IV
C. III, IV
D. IV, V

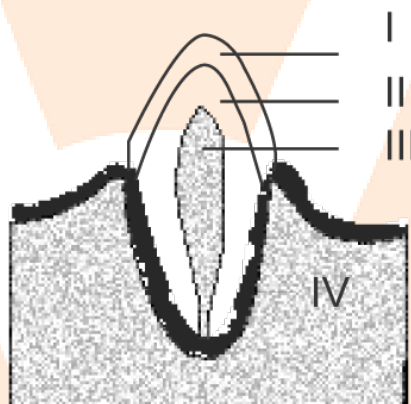
59. If 4 individuals are died and 8 individuals are born in a laboratory population of 40 fruit flies during time interval of 7 days. calculate the intrinsic rate of natural increase for fruit fly population per week
- 0.03
 - 3 0
 - 0.01
 - 0.1
60. Left side shift of oxygen-haemoglobin dissociation curve occurs during
- Low pH. low CO_2 . low temperature
 - High pH. low CO_2 , low temperature
 - High pH. High CO_2 , low temperature
 - Low pH, high CO_2 , high temperature
61. The following are the parts of human respiratory system
- Glottis
 - Trachea
 - External nostrils
 - Lain x
 - Nasal chambers
 - Bronchi
 - Alveolar ducts

VIII) Bronchioles

Arrange them in correct sequence through which inhaled air reaches the alveoli

- A. III, V, I, IV, II, VIII, VII
- B. VII, VIII, VI, II, IV, I, V, III
- C. V, I, IV, II, VI, VIII, VII, III
- D. III, V, I, IV, II, VI, VIII, VII

62. Study the following diagram of V.S. of tooth and identify the parts I. II. III and IV



- | | | | |
|------------|---------|-------------|-------------|
| I | II | III | IV |
| A. Dentine | Enamel | Pulp cavity | Periodontal |
| membrane | | | |
| I | II | III | IV |
| B. Enamel | Dentine | Pulp cavity | Periodontal |
| membrane | | | |
| I | | II | III |

C. Periodontal membrane Dentine Pulp cavity

IV

Enamel

I

II

III

D. Enamel Periodontal membrane Dentine

IV

Theca

63. Assertion (A) : Glomerular filtrate becomes hypertonic in distal convoluted tubule of the nephron
Reason (R) : In distal convoluted tubule obligatory reabsorption of water and Na^+ takes place

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

64. Study the following table

S. No.	Group	Number of chambers in heart	Type of circulation	Oxygenation of blood occurs in
I	Pisces	2	Single	Gills

II	Amphibia	3	Single	Skin, buccal cavity, lungs
III	Reptilia	3	Double	Lungs, skin
IV	Mammalia	4	Double	Lungs

Pick up the correct combination from the above

- A. I, IV
- B. II, II
- C. II, III
- D. III, IV

65. Study the following statements

- I) In the wall of veins of man elastic laminae are present on either side of tunica media
- II) Angina pectoris is due to insufficient supply of blood to heart muscles
- III) Juxta Glomerular cells of kidney secrete the enzyme rennin
- IV) Cellophane membrane of dialyser is impermeable to micromolecules like creatinine

Identify the incorrect statements among the above

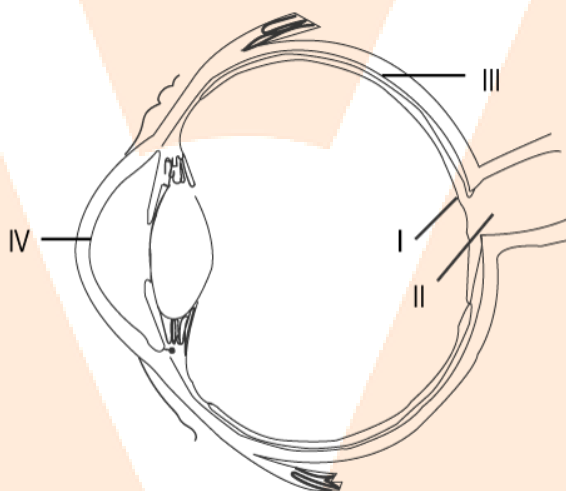
- A. I, II, III
- B. I, II

- C. I, III, IV
- D. II, III

66. Fibrous membrane in the middle of a sarcomere is

- A. A
- B. I
- C. Z
- D. M

67. Observe the following diagram of V.S. of eye and identify the parts I. II. III and IV



- | | I | II | III | IV |
|----|-------------|------------|-------------|---------|
| A. | Blind spot | Fovea | Choroid | Cornea |
| | I | II | III | IV |
| B. | Fovea | Blind spot | Choroid | Cornea |
| | I | II | III | IV |
| C. | Yellow spot | Fovea | Conjunctiva | Choroid |

I	II	III	IV
D. Blind spot	Yellow spot	Pupil	Cornea

68. Match the following

List – I

- A) Diabetes insipidus
- B) Diabetes mellitus
- C) Addison's disease
- D) Cushing's disease

List - II

- I) Cretinism
- II) Bronze coloured scars on skin
- III) Insulin
- IV) Buffalo hump
- V) Vasopressin

The correct answer is

A B C D

A. V III II IV

A B C D

B. V III I II

A B C D

C. V II III IV

A B C D

D. IV II III I

69. Study the following statements

- I) The largest endocrine gland of man secretes adrenalin

II) TH cells are involved in humoral and cell mediated immunity

III) Natural Killer cells are a type of lymphocytes

IV) Tc cells are involved in cellular immunity

Pick up the correct statements.

A. I, II

B. II, III

C. III, IV

D. I, IV

70. Acidity of urethra of man is neutralised by the secretion of

A. Prostate gland

B. Cowper's gland

C. Seminal vesicles

D. Skene glands

71. Study the following

s. No.	Foetal membranes in man	Formed by	Function
I	Yolk sac	Splanchnopleure	Not useful for nutrition
II	Allantois	Splanchnopleure	Store nitrogenous wastes

III	Amnion	Somatopleure	Fonn placenta
IV	Chorion	Splanchnopleure	Prevent dessication of embryo

Among the above, the correct combinations are

- A. I, II
- B. II, III
- C. III, IV
- D. I, IV

72. Pick up the mismatched pair

- A. Gonantoea - Neisseila
- B. Syphilis - Tireponema
- C. AIDS - HIV
- D. Cervix cancer - Thchomonas

73. A colour blind man married a woman who is a daughter of a colour blind father and normal homozygous mother. What are the chances of their daughters to become colour blind?

- A. 25 %
- B. 75 %
- C. 100 %

D. 50 %

74. Assertion (A) : Persons with O group of blood are called universal donors

Reason (R) : Their blood plasma contains both antigen A and antigen B

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

75. Study the following statements

I) Karyotype of Klinefelter's syndrome is $AA + XO$

II) Turner's syndrome is an example for trisomy

III) $AA + XX + 21\text{st chromosome}$ is the karyotype of Down's syndrome

IV) Individual with Klinefelter's syndrome is Barr body positive

Among the above, correct statements are

- A. I, II
- B. II, III
- C. III, IV
- D. I, IV

76. Which one of the following theories contradicts Lamark's theory?
- A. Theory of Natural Selection
 - B. Germplasm theory
 - C. Mutation theory
 - D. Theory of biogenesis
77. Assertion (A) : Long neck for Giraffes is due to directional selection
Reason (R) : It works constantly removing the individuals from one end and constantly shifting the average value to the other end.
- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
 - B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - C. (A) is correct but (R) is not correct
 - D. (A) is not correct but (R) is correct
78. Glue prepared and used by the worker bees to seal the cracks in honey combs
- A. Chrysalis
 - B. Propolis
 - C. Bee Wax

D. Bee Venom

79. Match the following

List – I

- A) Carcinoma
- B) Sarcoma
- C) Leukemia
- D) Lymphoma

List - II

- I) Cancer of connective tissue
- II) Cancer of muscular tissue
- III) Cancer of lymphatic system
- IV) Cancer of epithelial tissues
- V) Cancer of bone marrow

The correct answer is

- A B C D
- A. IV I II III
A B C D
 - B. IV I V III
A B C D
 - C. IV II V III
A B C D
 - D. III V I IV

80. Study the following statements

- I) Programmed death of cells is called apoptosis

- II) p⁵³ is called guardian angel of cell's genome
 III) Transgenic cow Rosie produced milk containing α -1 antitrypsin
 IV) Nitrosamines are non carcinogenic

Among the above, the incorrect statements are

- A. I, III
 B. I, II, III
 C. III, IV
 D. II, III, IV

81. If R , L , C , F , v , q , I and t represent resistance, inductance, capacitance, force, velocity, electric charge, electric current and time respectively, then which of the following will have same dimensions?

(a) $I^2 R$ (b) $\frac{L}{Rt}$ (c) $\frac{q^2}{RC^2}$ (d) $\frac{Fv}{t}$

- A. a & b
 B. a & c
 C. b & d
 D. a & d.

82. At time $t = 0$, a body is dropped freely from the top of a tall building and at a later time $t = T$, another body is thrown vertically downwards with a velocity ' v ' from the top of the same building. The time at which the two bodies will meet is

- A. $\frac{T}{2} \left[\frac{2v - gT}{v - gT} \right]$
- B. $\frac{T}{3} \left[\frac{3v - gT}{2v - gT} \right]$
- C. $\frac{T}{2} \left[\frac{v - gT}{2v - gT} \right]$
- D. $\frac{T}{3} \left[\frac{2v - gT}{3v - gT} \right]$

83. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of a projectile is 'p' times the maximum height attained by it, then the value of 'p' is

- A. $2\sqrt{3}$
- B. $3\sqrt{2}$
- C. $\sqrt{3}$
- D. $4\sqrt{3}$

84. Three particles A, B and C simultaneously start from the origin. Particle A moves with a velocity 'a' along X-axis. particle B moves with a velocity 'b' along Y-axis and particle C moves with a velocity 'c' in the X-

Y plane along the straight line $x = y$. The magnitude of 'c' so that all the three particles always remain collinear is

A. $a + b$

B. $\sqrt{ab}$

C. $\frac{ab}{a+b}$

D. $\frac{\sqrt{2}ab}{a+b}$

85. For a truck with 14 tyres, only rear 8 wheels are power driven and can produce acceleration. These 8 wheels support half the entire load. If the coefficient of friction between road and each tyre is 0.6. the maximum attainable acceleration by this truck would be (Acceleration due to gravity = 10 ms^{-2})

A. 6 ms^{-2}

B. 3 ms^{-2}

C. 10 ms^{-2}

D. 24 ms^{-2}

86. A soccer ball is travelling at a velocity 20 ms^{-1} due south. At the end of its travel, it moves with a velocity 2 ms^{-1} due south. If the change in the linear

momentum of the ball is 18 kgms^{-1} due north. then the mass of the ball is

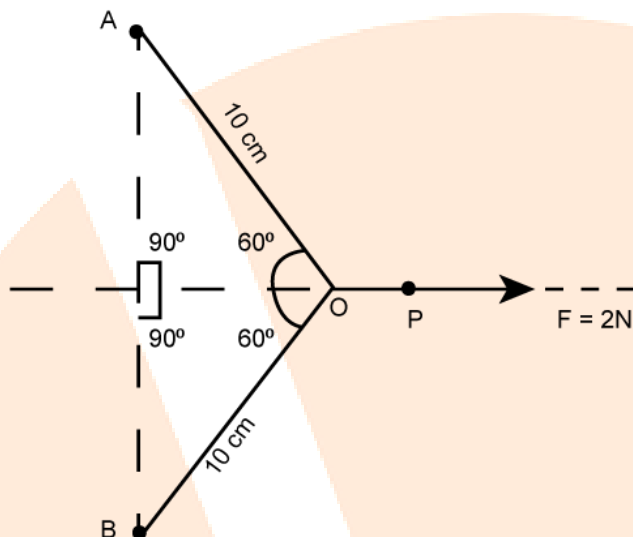
- A. 3.0 kg
- B. 0.81 kg
- C. 1.0 kg
- D. 0.5 kg

87. A stationary body explodes into four identical fragments such that three of them fly off E_0 mutually perpendicular to each other. each with a kinetic energy $\frac{E_0}{2}$. The total energy of - explosion is

- A. $6E_0$
- B. $3E_0$
- C. $\frac{3E_0}{2}$
- D. $\frac{2E_0}{3}$

88. Two point objects A and B each of mass 3 g and initially at rest are connected by massless threads as shown in the figure. A constant force 2 N acts on the system at point P along OP. After the point 'O' has moved a distance 20 cm in the direction of force. the two objects collide and stick together. The

displacement and velocity of the centre of mass of the system immediately after collision are respectively



- A. 20cm, 15 ms^{-1}
- B. 10cm, 15 ms^{-1}
- C. 15cm, 10 ms^{-1}
- D. 15cm, 20 ms^{-1}

89. A 13 m ladder is placed against a smooth vertical wall with its lower end at a distance 5 m from the wall. The minimum coefficient of friction between the ladder and the floor so that the ladder remains in equilibrium is nearly

- A. 0.52
- B. 0.72
- C. 0.21
- D. 0.36

90. Two blocks of masses 2 kg and 3 kg are attached with massless string passing over a fixed frictionless pulley as shown in the figure. When released, the velocity of the centre of mass of the system of two blocks after 1.5 seconds is (Acceleration due to gravity = 10 ms^{-2})
- 0.8 ms^{-1} upward
 - 0.6 ms^{-1} upward
 - 0.6 ms^{-1} downward
 - 0.8 ms^{-1} downward
91. A block of mass 1 kg tied to a long spring of spring constant 100 Nm^{-1} is at rest on a horizontal frictionless surface. The block is pulled through a distance 5 cm from its equilibrium position and released. Then the total energy of the block when it is at a distance 4 cm from the equilibrium position is
- 0.125 J
 - 12.5 J
 - 125 J
 - 1250 J
92. A number of planets are revolving around the Sun. Time period is 'T' and average orbital radius of a

planet is 'R'. A graph is drawn between $\log T$ on the Y-axis and $\log R$ on the X-axis with the origin at (0, 0). The graph is a

- A. 3 Straight line with slope $\frac{3}{2}$ and passing through the origin
- B. 3 Straight line with slope $\frac{3}{2}$ and not passing through the origin
- C. Parabola
- D. Ellipse

93. A wire is suspended vertically from a rigid support. When loaded with a body in air, the wire extends by 6 mm and when the body is immersed completely in water, the extension is reduced to 4 mm. The relative density of material of the body is

- A. $\frac{3}{2}$
- B. 3
- C. 2
- D. $\frac{2}{3}$

94. In a barometer, the mercury level is 76 cm at sea level. On a hill of height 3 km. if the ratio of density of Hg to that of air is 104, the atmospheric pressure on the hill is
- 26 cm of Hg
 - 46 cm of Hg
 - 36 cm of Hg
 - 56 cm of Hg
95. Two metal slabs of equal lengths. equal cross sectional areas and having resistances in the ratio 1 : 2 are connected first in series and then in parallel separately. The ratio of their effective conductivities is
- 1 : 2
 - 1 : 4
 - 4 : 9
 - 8 : 9
96. The ends of a uniform metal rod of length 100 cm and area of cross-section 2 cm^2 are maintained at 0°C and 100°C . At the mid point of the rod. heat is supplied at a constant rate of 40 Js^{-1} . If the temperature gradient on the higher temperature side of the rod in steady state is 50°Cm^{-1} . then the value

of x is (Thermal conductivity of the metal = $400 \text{ Js}^{-1} \text{ m}^{-1} \text{ K}^{-1}$)

- A. 2
- B. 3
- C. 6
- D. 9

97. The workdone on the system in changing the state of a gas adiabatically from equilibrium state A to equilibrium state B is 32.4 J. If the gas is taken from state A to B through another process in which the net heat absorbed by the system is 13.5 cal. then the net workdone by the system in the later case is (1 cal = 4.2 J)

- A. 32.4 J
- B. 56 J
- C. 14.3 J
- D. 24.3 J

98. A given mass of gas at a pressure ' P ' and absolute temperature ' T ' obeys the law $P \propto T^3$ during an adiabatic process. The adiabatic bulk modulus of the gas at a pressure ' P ' is

- A. $\frac{2P}{3}$

- B. P
- C. $\frac{3P}{2}$
- D. $2P$

99. The number of degrees of freedom of a gas whose specific heat capacity at constant pressure is $33.24 \text{ J mol}^{-1} \text{ K}^{-1}$, is (universal gas constant = $8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

- A. 2
- B. 3
- C. 6
- D. 8

100. A man standing far from a hill. fires a gun and hears its echo after 4 s. Later he moves 320 m from his initial position away from the hill and fires the gun again and now he hears the echo after 6 s. Then the velocity of the sound in air is

- A. 330 ms^{-1}
- B. 340 ms^{-1}
- C. 320 ms^{-1}
- D. 336 ms^{-1}

101. A whistle of frequency 540 Hz rotates along a circle of radius 2 in at an angular speed of 15 rad s^{-1} . The

difference in maximum and minimum frequencies heard by a listener at some distance and at rest with respect to the centre of the circle. is (Speed of sound in air = 330 ms^{-1})

- A. 99 Hz
- B. 59 Hz
- C. 49 Hz
- D. 109 Hz

102. Match the following List-I with the List-II

List - I List - II

- A) Mirages I) Semiconductor
- B) Nichrome II) Natural convection
- C) Germanium III) Total internal reflection
- D) Sea breeze IV) High resistivity

A B C D

- A. III IV I II

A B C D

- B. IV III II I

A B C D

- C. II III I IV

A B C D

- D. I II III IV

103. A beam of light is incident on a glass slab of refractive index 1.54 in a direction as shown in the figure. The reflected light is analysed by a polaroid prism. On rotating the polaroid ($\tan 57^\circ = 1.54$)
- The intensity remains unchanged
 - The intensity is reduced to zero and remains at zero
 - The intensity gradually reduces to zero and then increases
 - The intensity increases gradually
104. A solid conducting sphere of radius 20 cm is enclosed by a thin metallic shell of radius 40 cm. A charge of 40 pLC is given to the inner sphere. If the metallic shell is earthed, then the heat generated in the process is
- 18J
 - 9J
 - 36 J
 - 1.8 J
105. A particle of mass 1 kg carrying a charge of 0.01 C is able to remain at rest on a rough inclined plane of inclination 30° when a uniform horizontal electric

field of $\frac{490}{\sqrt{3}} Vm^{-1}$ is applied. Coefficient of friction is

(Acceleration due to gravity = 9.8 ms^{-2})

A. 0.5

B. $\frac{1}{\sqrt{3}}$

C. $\frac{\sqrt{3}}{7}$

D. $\frac{\sqrt{3}}{2}$

106. Two identical rings each of radius 'R' are coaxially placed a distance 'R' apart. They carry charges Q_1 and Q_2 respectively. If a charge 'q' is moved from the centre of one ring to the centre of the other ring. the work done is

A. 0

B. $\frac{q(Q_1 - Q_2)(\sqrt{2} - 1)}{\sqrt{2}(4\pi \epsilon_0 R)}$

C. $\frac{q\sqrt{2}(Q_1 + Q_2)}{4\pi \epsilon_0 R}$

D.
$$\frac{q(Q_1 + Q_2)}{(\sqrt{2} + 1)4\pi \epsilon_0 R}$$

107. A conducting spherical bubble of radius 'r' and thickness t ($t \ll r$) is charged to a potential 'V'. If it collapses to form a small spherical drop, then the potential of the drop is

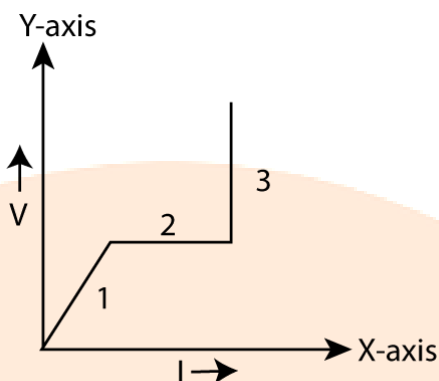
A. $V(3tr)^{\frac{1}{3}}$

B. $\left(\frac{3t}{r}\right)^{\frac{1}{3}} V$

C. $\left(\frac{r}{3t}\right)^{\frac{1}{3}} V$

D. $\left(\frac{3Vt}{r}\right)^{\frac{1}{3}}$

108. A graph drawn between current I and voltage 'V' in a conductor is as shown in the figure. The changes in the resistance in 1st and 3rd parts respectively



- A. Remains constant. Increases
- B. Decreases. Increases
- C. Increases, remains constant
- D. Decreases. remains constant

109. A steady current is passing through a cylindrical conductor of radius 'r' placed in vacuum. Assuming Stefan's law of radiation, steady temperature will be proportional to

- A. r^{-3}
- B. $r^{\frac{3}{4}}$
- C. $r^{-\frac{2}{3}}$
- D. $r^{-\frac{3}{5}}$

110. The two ends of a non-conducting spring of force constant 50 N m^{-1} and unstretched length of 2 cm are connected to the mid points of two straight parallel

rods each of length 4 in. When 100 A current is passed through each rod in the same direction. the work done on the spring in mJ is

- A. 0.8
- B. 1.6
- C. 3.2
- D. 6.4

111. The magnetic field normal to the plane of a coil of 'n' turns and radius 'r' carrying a current 'i' is measured on the axis of the coil at a distance 'h' ($h \ll r$) from the centre of the coil. This is smaller than the field at the centre by the fraction

- A. $\frac{2h^2}{3r^2}$
- B. $\frac{3r^2}{2h^2}$
- C. $\frac{3h^2}{2r^2}$
- D. $\frac{2r^2}{3h^2}$

112. A thin bar magnet oscillates with a time period T. If it is cut into two equal pieces along its axis, time period of oscillation of each piece is

- A. T
- B. 2T
- C. $\frac{T}{2}$
- D. $\frac{T}{4}$

113. A coil having 200 turns is placed in a magnetic field of $60 e^{-200t}$ T with its plane perpendicular to the magnetic field. The cross sectional area of the coil is $5e^2 \text{ cm}^2$. The ends of the coil are connected to 100Ω resistance. The current in the coil at $t = 0.01$ second is

- A. 6A
- B. 12A
- C. 18A
- D. 24A

114. A steady current of 2A flows when an inductor of inductance 2 mH is connected to ac source of emf 10 V. Now a capacitor of capacity $2\mu\text{F}$ is connected in series. If the current in the circuit is along the emf. the rms value of the current is

- A. 2.0A
- B. 1.5A
- C. 5.0A

D. 0.2A

115. A parallel plate capacitor with plate area 1 m^2 and plate separation 1 mm is charged at the rate of 25 Vs^{-1} . The dielectric between the plates has a dielectric constant 'k'. If the displacement current through the capacitor is $2.21 \mu\text{A}$ then the value of 'k' is nearly

- A. 25
- B. 20
- C. 15
- D. 10

116. In a photoelectric effect experiment, collector plate is placed vertically above the emitter plate. Light is allowed to incident on emitter and saturation photocurrent is recorded. Now parallel electric and magnetic fields are applied vertically downwards between the plates. Then

- A. The photocurrent will increase
- B. The kinetic energy of photoelectrons will decrease
- C. The stopping potential will increase
- D. The threshold wavelength will increase

117. Hydrogen atom in its ground state is excited by means of monochromatic radiation of energy 12.75 eV. The minimum energy of the emitted spectral lines is nearly
- 0.22 eV
 - 0.44 eV
 - 0.66 eV
 - 0.88 eV
118. Two radio active nuclei 'x' and 'y' initially contain equal number of atoms. Their half-life periods are 1 hour and 2 hours respectively. The ratio of their rates of disintegration after 2 hours from the start is
- 1 : 1
 - 2 : 1
 - 2 : 1
 - 1 : 3
119. The current gain of a transistor in a common emitter circuit is 25. then the ratio of emitter current to base current is
- $\frac{25}{26}$
 - 26
 - $\frac{1}{25}$

D. $\frac{26}{25}$

120. Assertion (A) : FM signal is less susceptible to noise than M signal.

Reason (R) : FM has small operating frequency range.

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

121. The kinetic energy of electron ejected from a metal surface is 0.70 eV. If the work function (W_0) of the metal is 2.30 eV, the frequency of radiation falling on the metal surface in Hz is ($1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$)

- A. 7.25×10^{13}
- B. 1.38×10^{13}
- C. 1.38×10^{14}
- D. 7.25×10^{14}

122. If the ratio of radii of electron in the first excited states of He and Be^{3+} is 2:1, the energies of electron in their corresponding excited states in J is

- A. $-2.18 \times 10^{-18}, -8.72 \times 10^{-18}$
- B. $-8.72 \times 10^{-18}, -2.18 \times 10^{-18}$
- C. $-2.18 \times 10^{-18}, -4.36 \times 10^{-18}$
- D. $-4.36 \times 10^{-18}, -2.18 \times 10^{-18}$

123. Identify the correct statements from the following

- i. CO is an amphoteric oxide
- ii. As_2O_3 is a neutral oxide
- iii. Cl_2O_7 is an acidic oxide
- iv. CaO is a basic oxide

- A. i, ii, iii, iv
- B. ii, iii, iv only
- C. iii, iv only
- D. iv. ii only

124. Atom X (in molecule 1) has electrons in 1 s. 2s and 2p orbitals. Atom Y (in molecule 2) has electrons in 1s. 2s. 2p. 3s and 3p orbitals. Atom Z (in molecule 3) has electrons in 1s. 2s, 2p. 3s. 3p. 3d and 4s orbitals. Observe the following statements

- i. Atom X can undergo sp , sp^2 and sp^3 type of hybridisation
- ii. Atom Y can undergo sp^3 , sp^3d type of hybridisation
- iii. Atom Z can undergo sp^3d^2 , d^2sp^3 , dsp^2 , sp^3 type of hybridisation

The correct statements are

- A. i, ii only
- B. i, iii only
- C. ii, iii only
- D. I, ii, iii

125. The number of electrons present in valency shell of S in SF_6 , H_2SO_4 and SCl_2 are respectively

- A. 12, 12, 6
- B. 12, 12, 4
- C. 12, 8, 12
- D. 12, 12, 8

126. A hydrogen, neon gas mixture contains 0.5 g of hydrogen and 10 g of neon in a cylinder. If pressure of the mixture of gases in the cylinder is 2.4 bar, the partial pressures of neon and hydrogen in the mixture respectively in bar are ($H = 1.0$; $Ne = 20.0$)

- A. 0.9, 1.5
- B. 1.6, 0.8

C. 0.8 , 1.6

D. 1.5. 0.9

127. 20 mL of 0.04M CuSO_4 solution reacts completely with 20 mL of certain concentration of KI solution. If 12 thus liberated reacts completely with 20 mL of sodium thiosulphate solution, the molarity of thiosulphate in mol L^{-1} is

A. 0.01

B. 0.02

C. 0.04

D. 0.08

128. The enthalpy and entropy changes for a reaction at 298 K are 400 kJ mol^{-1} . 400 J mol^{-1} respectively. If ΔH and ΔS are constant over the temperature range. the temperature at which the reaction becomes spontaneous in K is

A. 1001

B. 800

C. 101

D. 500

129. If the equilibrium constant, K_c for a reaction at certain temperature is $> 10^3$, which one of the following statements is correct?
- A. The reaction hardly proceeds
 - B. The products predominate over reactants at equilibrium
 - C. The reactants predominate over products at equilibrium
 - D. The equilibrium constant is less than reaction quotient

130. Assertion (A) : Higher order ionization constants (K_{a2} , K_{a3}) are smaller than the lower order ionization constant (K_a) of polyprotic acid
Reason (R) : It is easy to remove a $1.1+$ from a negative ion due to electrostatic forces

The correct answer is

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (B) are correct but (R) is not the correct explanation of (A)
- C. (A) is correct but (R) is not correct
- D. (A) is not correct but (R) is correct

131. What is the approximate volume (in L) of 0.02 M KMnO_4 solution required to completely react with 0.5 L of 10 vol H_2O_2 solution in acid medium?
- 89.29
 - 44.64
 - 4.464
 - 8.929
132. In portland cement. the major constituents (in mol %) are
- MgO , Fe_2O_3 , Cr_2O_3
 - Al_2O_3 , MgO , TiO_2
 - CaO , SiO_2 , Al_2O_3
 - Al_2O_3 B_2O_3 , MgO
133. Sodium borohydride reacts with I_2 to form NaI , H_2 , and X. When X reacts with NaH in diethyl ether. Y is formed. What is Y?
- B_2H_6
 - H_3BO_3
 - B_2O_3
 - NaBH_4
134. Identify the correct statements from the following
- $\Delta_f H^\circ$ of graphite is taken as zero

- ii. In fiderene. every five carbon ring is fused to six carbon rings only
- iii. CO_2 is prepared by dehydrating formic acid using conc. H_2SO_4 at 373 K
- A. i, ii, iii
- B. ii, iii only
- C. i, ii only
- D. ii, iii only

135. Identify correct statements from the following

- i. Conventional smog is a mixture of smoke. SO_3 and fog
- ii. Conventional smog has oxidizing nature
- iii. In the photochemical smog. the natant unsaturated hydrocarbons are converted into PAN. acrolein and HCHO
- iv. Photochemical smog can be controlled by fixing catalytic converters in automobile engines
- A. i, ii, iii, iv
- B. i, ii, ii only
- C. iii, iv only
- D. i, ii only

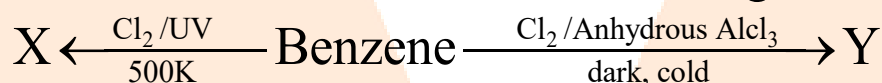
136. In C and H estimation experiment. H_2O and CO_2 formed are absorbed in A and B respectively. A and B are

A	B
A. anhydrous CuSO_4	Ca(OH)_2 solution
B. anhydrous CuSO_4	KOH solution
C. anhydrous CaCl_2	KOH solution
D. anhydrous CaCl_2	Ca(OH)_2 solution

137. Ozonolysis of alkene X gives acetone. X is

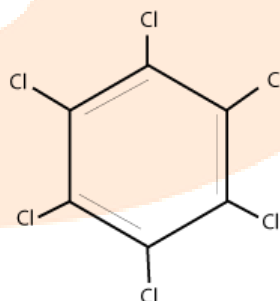
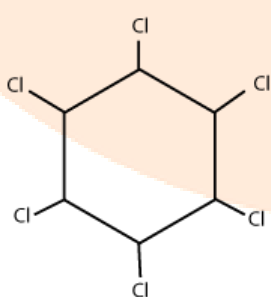
- A. 2, 3 - diethyl but - 2 - ene
- B. 2 - Methylpropene
- C. but - 2 - ene
- D. 2 - Methyl but - 2 - ene

138. What are X and Y in the following reactions ?

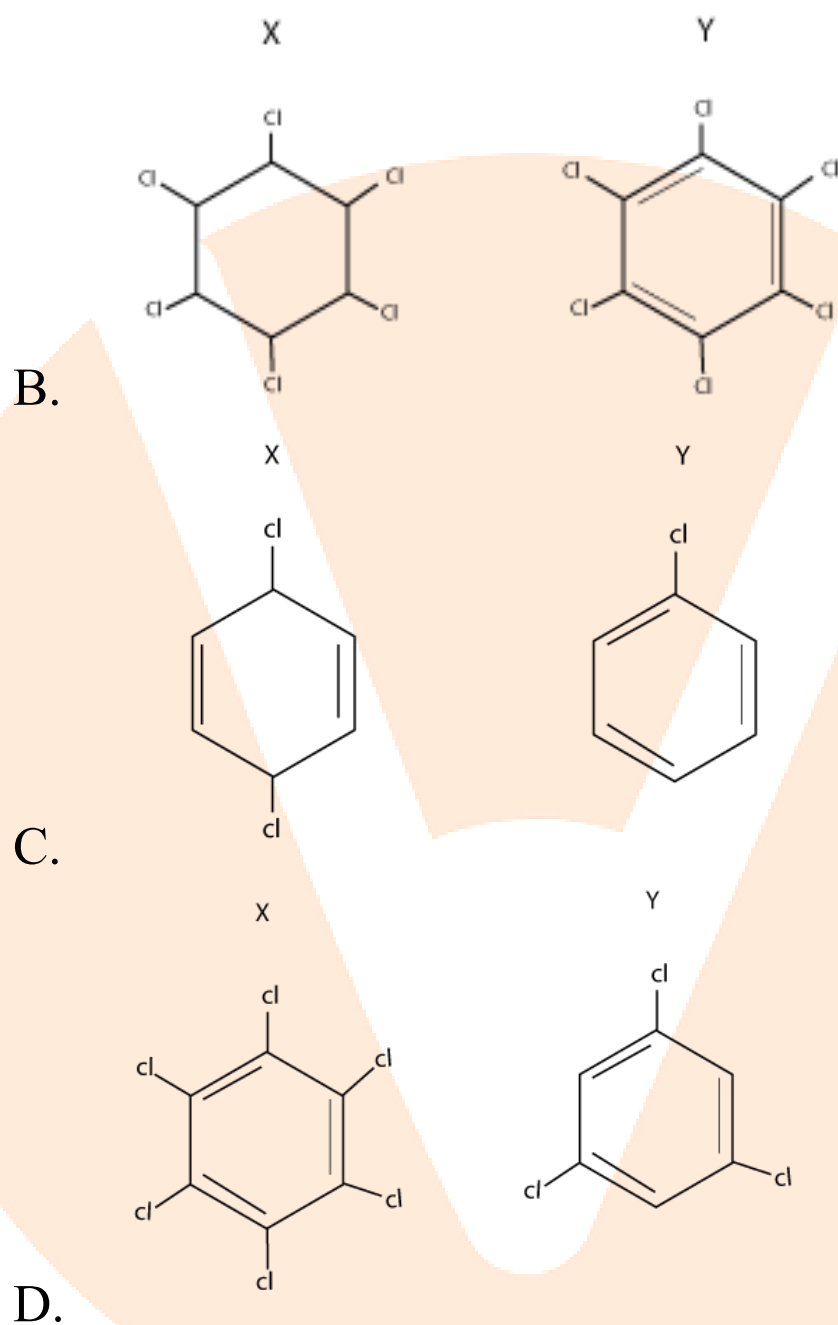


X

Y



A.



139. An element has body-centered cubic structure with a cell edge length of 300 pm. If the density of the

element is 7.2 g cm^{-3} . the number of unit cells in 194.4 g of the element is

- A. 2.0×10^{24}
- B. 6.0×10^{24}
- C. 3.0×10^{24}
- D. 1.0×10^{24}

140. The vapour pressure of a solution of 6.0 g of non-volatile solute in 390 g of benzene at 298 K is 3.00 kPa . If 78 g of benzene is added to this solution the vapour pressure becomes 3.02 kPa at the same temperature. The molar mass of solute in g mol^{-1} is

- A. 60.8
- B. 50.4
- C. 31.2
- D. 21.2

141. The mole fractions of benzene and toluene vapours in equilibrium with the ideal solution of benzene in toluene at 300 K are 0.61 and 0.39 respectively. The total vapour pressure of the solution is 41 mm Hg . If the vapour pressures of pure benzene and toluene at 300 K are 50 and 32 mm Hg respectively, the mole fractions of benzene and toluene in solution respectively are

- A. 0.25, 0.75
- B. 0.75, 0.25
- C. 0.30, 0.70
- D. 0.50, 0.50

142. Solutions of CuSO_4 and AgNO_3 were electrolysed with a current of 1.93 amperes for 500 seconds separately. The amounts of copper and silver deposited at cathode respectively in g are

- A. 0.63, 0.54
- B. 0.315, 0.54
- C. 0.315, 1.05
- D. 1.08, 0.315

143. If the rate constant of zero order reaction is $3.0 \times 10^{-3} \text{ Ms}^{-1}$, the time taken for the initial concentration of the reactant to fall from 0.30 M to 0.03 M in seconds is

- A. 90
- B. 10
- C. 60
- D. 30

144. Which one of the following methods is used to make gold so!?

- A. Ultra filtration method
- B. Peptization method
- C. Electrical disintegration method
- D. Dialysis method

145. In the van-Arkel method of refining Zr. the impurities (x) and the substance (y) used are respectively

- A. $x = (N_2, O_2); y = Cl_2$
- B. $x = (N_2, O_2); y = O_2$
- C. $x = (N_2, O_2); y = I_2$
- D. $x = (SO_2, O_2); y = I_2$

146. An oxoacid of P(A) on heating disproportionates to form an acid (X) and a base (Y). Y reacts with $CuSO_4$ solution to form Cu_3P_2 and sulphuric acid. A is a dibasic acid. A. X and Y are respectively

- A. H_3PO_4 , H_3PO_3 , PH_3
- B. H_3PO_2 , H_3PO_4 , PH_3
- C. H_3PO_3 , PH_3 , H_3PO_4
- D. H_3PO_3 , H_3PO_4 , PH_3

147. Phosphorous reacts with sulphuryl chloride to form PCl_5 and X. Hot and concentrated H_2SO_4 reacts with

Cu to form CuSO_4 , H_2O and Y. Zinc sulphide reacts with oxygen to form ZnO and Z. Then X, Y and Z are respectively

- A. SO_2 , SO_2 , H_2S
- B. SO_2 , SO_2 , SO_2
- C. SO_2 , H_2S , SO_2
- D. SO_3 , SO_2 , SO_3

148. Which one of the following diffuses through rubber, glass and plastic?

- A. He
- B. Ne
- C. Ar
- D. Kr

149. Which pair of elements does not exhibit multiple oxidation states?

- A. Mn, Cr
- B. Sc, Zn
- C. Fe, Co
- D. V, Ni

150. $[\text{ML}_6]^{n+}$ is an octahedral complex. Its crystal field splitting energy (Δ_0) is 1.8 eV. If same metal ion (M^{n+}) forms a tetrahedral complex with same ligands

(L), then the crystal field splitting energy (Δ_t) of $[\text{ML}_4]^{n+}$ in eV is

- A. 1.8
- B. 0.8
- C. 4.05
- D. 0.9

151. Identify homopolymers from the following
nylon6 orlon bakelite glyptal neoprene

I II III IV V

- A. II, III, IV
- B. I, II, V
- C. I, III, IV
- D. II, IV, V

152. Examples of fibrous (X) and globular (Y) proteins are

- | | |
|--------------|-----------|
| X | Y |
| A. Insulin | albumin |
| X | Y |
| B. Keratin | albumin |
| X | Y |
| C. Keratin | thyroxine |
| X | Y |
| D. thyroxine | insulin |

153. Match the following.

List - I

- A) Receptor
- B) Bactericidal
- C) Neurotransmitter
- D) Bacteriostatic

List - II

- I) Noradrenaline
- II) Tetracycline
- III) Protein
- IV) Ranitidine
- V) Penicillin

The correct answer is

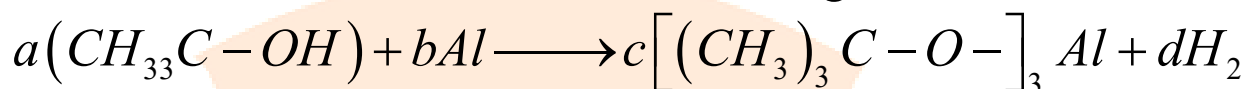
- (A) (B) (C) (D)
- A. I V III II
- (A) (B) (C) (D)
- B. III V I II
- (A) (B) (C) (D)
- C. III IV V II
- (A) (B) (C) (D)
- D. IV II I V

154. The order of reactivity of chlorobenzene (I), 4 - nitrochlorobenzene (II), 2,4, 6 - trinitrochlorobenzene (III) and 2, 4 - dinitrochlorobenzene (IV) towards nucleophilic substitution is

- A. III > II > IV > I
- B. III > IV > II > I
- C. II > III > IV > I

D. $I > II > IV > III$

155. What are a. b. c and d in the following reaction?



A. $\frac{a}{6} \frac{b}{2} \frac{c}{2} \frac{d}{3}$

B. $\frac{a}{3} \frac{b}{1} \frac{c}{1} \frac{d}{2}$

C. $\frac{a}{4} \frac{b}{1} \frac{c}{1} \frac{d}{2}$

D. $\frac{a}{4} \frac{b}{2} \frac{c}{1} \frac{d}{2}$

156. Lucas reagent is

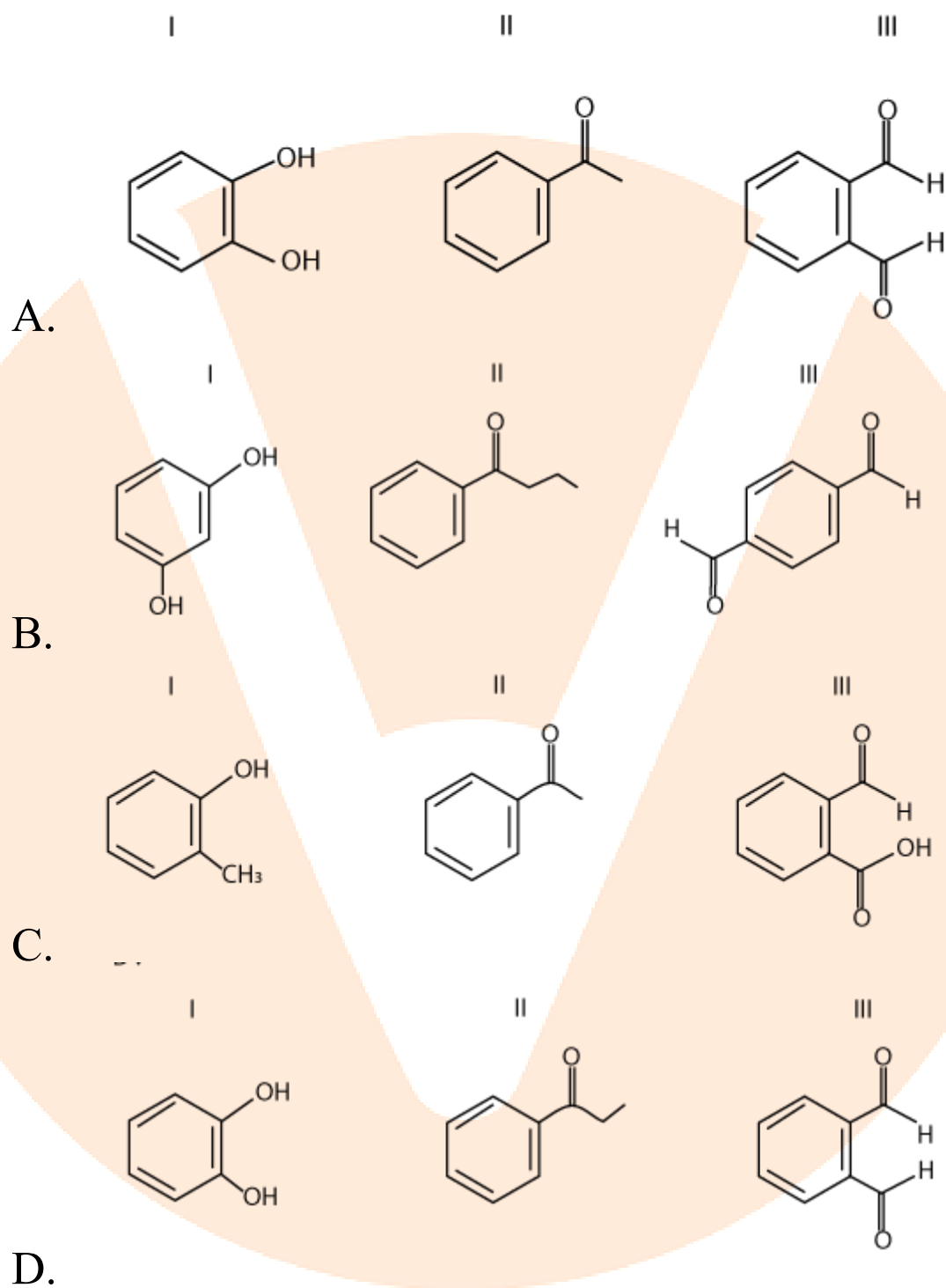
A. Br_2/CCl_4

B. $KMnO_4 / H^{\oplus}$

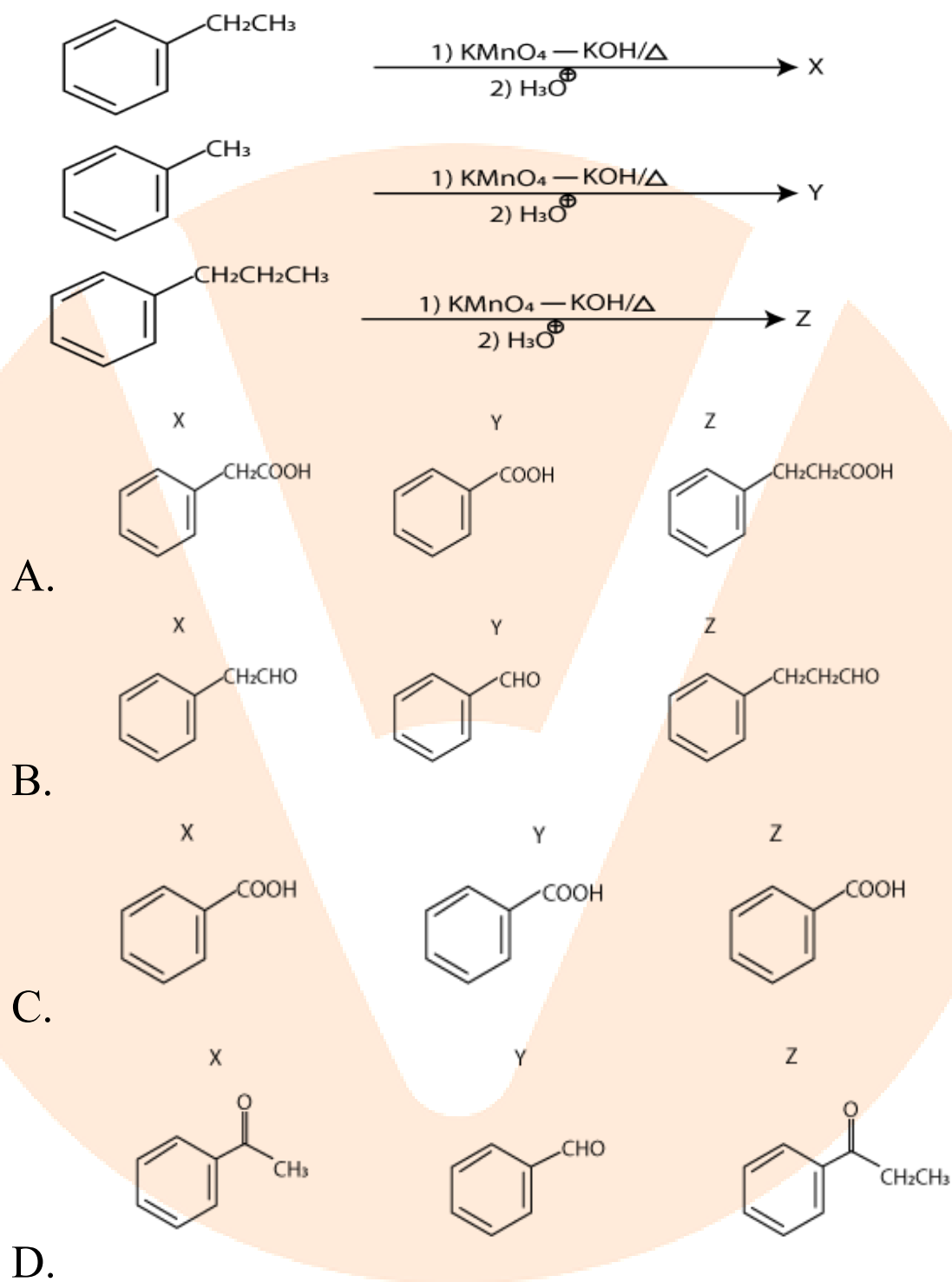
C. $HCl / \text{anhydrous } ZnCl_2$

D. $C_6H_5SO_2Cl$

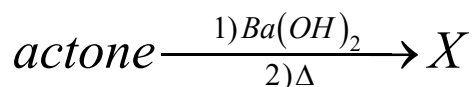
157. Structures of catechol propiophenone (II) and phthalaldehyde (III) are



158. What are X, Y and Z in the following reactions ?



159. Identify X in the following reaction



- A. 4 - methylpent - 3 - ene - 2 - one
- B. 3 - methylpent - 3 - ene - 2 - one
- C. hex - 3 - ene - 2 - one
- D. pent - 3 - ene - 2 - one

160. What are X and Y in the following reactions?

